

Below are 100 viva/practice questions and answers specifically for your MindAssist-AI: AI-Based Mental Health Research Assistant project.

1. What is MindAssist-AI?

Answer: MindAssist-AI is an AI-powered mental health research assistant that uses Agentic AI and Retrieval-Augmented Generation (RAG) to answer mental health research questions based on trusted PDF documents.

2. What is the main objective of your project?

Answer: To provide accurate, evidence-based mental health information using AI while reducing hallucinations.

3. Why did you choose the mental health domain?

Answer: Mental health problems are increasing among university students, and reliable information is essential.

4. What problem does your system solve?

Answer: It helps users quickly access reliable mental health research information from scientific papers.

5. What is Agentic AI?

Answer: Agentic AI is an AI system that uses multiple intelligent agents to complete tasks collaboratively.

6. What is RAG?

Answer: Retrieval-Augmented Generation retrieves relevant documents before generating an answer.

7. Why is RAG better than a normal chatbot?

Answer: RAG generates answers from trusted documents instead of relying only on the LLM's knowledge.

8. What is hallucination in AI?

Answer: Hallucination is when an AI generates false or unsupported information.

9. How does your project reduce hallucinations?

Answer: By retrieving relevant document chunks and validating answers using a Reflection Agent.

10. Which programming language did you use?

Answer: Python.

11. Which framework did you use for the UI?

Answer: Streamlit.

12. Which vector database did you use?

Answer: ChromaDB.

13. Which embedding model did you use?

Answer: all-MiniLM-L6-v2.

14. Why did you choose ChromaDB?

Answer: It is free, lightweight, open-source, and easy to integrate with LangChain.

15. Which LLM did you use?

Answer: Groq Llama 3.1 8B for routing and OpenRouter models for reasoning.

16. Why use Groq?

Answer: Groq provides very fast inference with low latency.

17. Why use OpenRouter?

Answer: It provides access to multiple advanced LLMs.

18. What is LangChain?

Answer: LangChain is a framework for building LLM applications.

19. What is LangGraph?

Answer: LangGraph is a framework for creating multi-agent workflows.

20. What is a Router Agent?

Answer: It classifies the user's request and selects the correct workflow.

21. What is an Orchestrator Agent?

Answer: It coordinates all other agents.

22. What is a Research Agent?

Answer: It retrieves information from the vector database.

23. What is a Reflection Agent?

Answer: It verifies the generated answer before showing it to the user.

24. Why is the Reflection Agent important?

Answer: It improves reliability and reduces hallucinations.

25. What is Tool Use?

Answer: An AI agent using external tools like databases or APIs.

26. How are PDFs loaded?

Answer: Using PyPDFDirectoryLoader.

27. Why split documents?

Answer: Large documents cannot be embedded directly.

28. What chunk size did you use?

Answer: 1000 characters.

29. What overlap did you use?

Answer: 200 characters.

30. Why use chunk overlap?

Answer: To preserve context between chunks.

31. What is an embedding?

Answer: A numerical representation of text.

32. Why are embeddings important?

Answer: They enable semantic search.

33. What is semantic search?

Answer: Searching based on meaning instead of keywords.

34. What is cosine similarity?

Answer: A metric for measuring similarity between vectors.

35. How does ChromaDB search?

Answer: It finds the most similar embeddings.

36. What is top-k retrieval?

Answer: Retrieving the top K most relevant chunks.

37. What value of K did you use?

Answer: Five.

38. Why retrieve multiple chunks?

Answer: To provide better context.

39. What is prompt engineering?

Answer: Designing prompts to improve AI responses.

40. What is context window?

Answer: Maximum tokens an LLM can process.

41. What is Streamlit?

Answer: A Python framework for web applications.

42. Why use Streamlit?

Answer: It is simple and fast.

43. What is virtual environment?

Answer: An isolated Python environment.

44. What is requirements.txt?

Answer: It lists project dependencies.

45. Why use .gitignore?

Answer: To exclude unnecessary files.

46. Why hide API keys?

Answer: For security.

47. Where are API keys stored?

Answer: Streamlit Secrets.

48. What is Git?

Answer: A version control system.

49. What is GitHub?

Answer: A cloud platform for Git repositories.

50. Why use GitHub?

Answer: Collaboration and version control.

51. What is a commit?

Answer: A saved version of code.

52. What is branching?

Answer: Creating separate development paths.

53. What is merging?

Answer: Combining branches.

54. What is deployment?

Answer: Publishing the application online.

55. Where is your project deployed?

Answer: Streamlit Community Cloud.

56. What is a PDF loader?

Answer: It extracts text from PDF files.

57. What is RecursiveCharacterTextSplitter?

Answer: It splits documents into chunks.

58. Why use HuggingFace embeddings?

Answer: They are free and accurate.

59. What is vector search?

Answer: Searching using embeddings.

60. What is similarity search?

Answer: Finding related documents using vector distance.

61. What is JSON?

Answer: A lightweight data format.

62. How do agents communicate?

Answer: Using structured JSON messages.

63. What is latency?

Answer: Time taken to generate a response.

64. Why is low latency important?

Answer: It improves user experience.

65. What is accuracy?

Answer: Correct predictions over total predictions.

66. What is precision?

Answer: Correct positive predictions over predicted positives.

67. What is recall?

Answer: Correct positives over actual positives.

68. Why is recall important in mental health?

Answer: Missing a depressed person is dangerous.

69. What is F1-score?

Answer: Harmonic mean of precision and recall.

70. What is overfitting?

Answer: Learning training data too closely.

71. What is underfitting?

Answer: Model fails to learn patterns.

72. What datasets are used?

Answer: Mental health research papers and PDFs.

73. Why use scientific papers?

Answer: They are reliable.

74. Can your chatbot diagnose depression?

Answer: No. It provides research-based information only.

75. Does it replace a doctor?

Answer: No.

76. What are ethical issues?

Answer: Privacy, bias, transparency, and safety.

77. How is user privacy protected?

Answer: API keys are secured and no unnecessary personal data is stored.

78. What is bias?

Answer: Unfair predictions affecting some groups.

79. How can bias be reduced?

Answer: Using diverse and validated datasets.

80. What are future improvements?

Answer: Voice support, multilingual support, and memory.

81. Can images be analyzed?

Answer: Not in the current version.

82. Can web search be added?

Answer: Yes.

83. Can multiple agents work together?

Answer: Yes.

84. What is orchestration?

Answer: Managing multiple agents.

85. Why use multiple agents?

Answer: Better specialization and accuracy.

86. What is a workflow?

Answer: Sequence of agent tasks.

87. What happens if retrieval fails?

Answer: Reflection Agent informs the user that information is insufficient.

88. What happens if API fails?

Answer: Error handling displays a friendly message.

89. What are project limitations?

Answer: Depends on uploaded PDFs and internet APIs.

90. What makes your project unique?

Answer: Multi-agent architecture with validated RAG.

91. What software was used?

Answer: Python, Streamlit, LangChain, ChromaDB, Groq, OpenRouter, GitHub.

92. What operating system did you use?

Answer: Windows 11.

93. Why use Python?

Answer: Rich AI ecosystem and libraries.

94. How many AI agents are used?

Answer: Four.

95. Name the four agents.

Answer: Router, Orchestrator, Research, Reflection.

96. What is the biggest advantage of your project?

Answer: Reliable answers from trusted research documents.

97. What is the biggest challenge?

Answer: Preventing hallucinations.

98. How do you test the system?

Answer: By asking sample research questions and checking retrieved documents.

99. Who are the target users?

Answer: Students, researchers, and educators interested in mental health.

100. Summarize your project in one sentence.

Answer: MindAssist-AI is a multi-agent AI research assistant that uses Retrieval-Augmented Generation (RAG) to provide accurate, evidence-based mental health information from trusted research papers.